Claim, Evidence, and Falsification

in a Mathematical-Physics Compendium

A critical review of algebraic constructions, computational artifacts, and speculative physical mappings

Eirikr Hinngart

Independent researcher; MathScienceCompendium project

Correspondence: `https://github.com/Oichkatzelesfrettschen/MathScienceCompendium/issues`

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Abstract

This review evaluates a repository of mathematical-physics claims against three distinct standards: theorem-level mathematical support, reproducible computation, and empirical physical evidence. The underlying corpus combines established structures, including exceptional Lie algebras, Cayley–Dickson algebras, modular forms, and fractal dimensions, with proposed mappings to force unification, quantum encoding, materials, and two-dimensional turbulence. The mathematical objects are often standard; the proposed physical maps generally are not consequences of those objects.

The audit therefore separates identity from mechanism and computation from experiment. In particular, the identity c^4/G defines a force scale but does not by itself unify interactions. Likewise, the determinant of a Cartan matrix does not induce a beta-plane parameter unless a mode map, dynamical equation, and controlled comparison establish that mechanism. The review converts every claim on the bounded canonical ledger into an executable consistency contract and every bounded open claim into a separate hypothesis record. Contract consistency is not scientific admission. The review repairs the E_7 and extended Cartan constructions, implements a nontrivial Albert-algebra product, exhibits exact higher Cayley–Dickson failure witnesses, and closes the external Fourier-to-P/Q filter negatively. The proposed quadratic-defect replacement is exactly a generic checkerboard kernel at radii 2, 4, and 8, so the registered E_7-specificity hypothesis fails. A preregistered 540-run barotropic beta-plane program likewise does not support its locked transient-zonalization hypothesis: all four required $\beta = 5, 10$ contrasts miss their simultaneous confidence and minimum-effect gates. A cache-isolated rerun in a clean pinned container reproduces both negative decisions without primary-cache reuse. These results apply to the declared freely decaying model and selector, not to forced–dissipative jet formation or to every parity-kernel mechanism. Physical admission packages retain conventional tourmaline and time-crystal phenomena but reject excess-energy conclusions without complete energy accounting. The resulting manuscript is a claim audit, negative-result record, and reproducibility map, not a unified-field theory.

Contents

CHAPTER 1

Scope, Questions, and Review Method

1.1 What this manuscript reviews

The repository combines three artifact classes that require different tests:

1. established mathematical constructions and their implementations;
2. numerical experiments that test algorithms or dynamical models; and
3. physical interpretations proposed in source manuscripts.

The classes are related, but they are not interchangeable. A correct Cartan matrix does not validate a fluid model. A stable numerical trajectory does not prove a theorem. A dimensional identity does not establish a new interaction. The review treats every transition between classes as a separate claim that needs its own argument and evidence.

1.2 Questions

The audit asks four questions.

Definition
: Is the mathematical or physical object defined well enough to determine what would count as an error?

Derivation
: Does the conclusion follow from the premises, with all additional assumptions stated?

Reproduction
: Can the reported computation be regenerated from a named program, parameter set, and retained output?

Falsification
: What observation or controlled comparison would reject the proposed physical interpretation?

These questions are intentionally conservative. They do not ask whether a cross-domain analogy is interesting. They ask whether the analogy has been promoted to a mechanism without the work needed to justify that promotion.

1.3 Evidence vocabulary

The earlier manuscript used terms such as “verified”, “proved”, and “confirmed” for several incompatible kinds of support. This edition uses the following controlled vocabulary.

Table 1.1: Evidence classes used throughout the review.

Class	Required support
Established result	A theorem or standard result with a suitable source.
Source claim	A statement found in the reviewed corpus, with no implied endorsement.
Computational check	A program and retained output that test a specified finite case.
Hypothesis	A proposed relation with a stated test and possible failure outcome.
Unsupported extrapolation	A conclusion for which the repository supplies no valid derivation or discriminating evidence.
Invalid artifact	An output that is malformed or fails a prerequisite numerical check.

1.4 Evidence surfaces

The review uses repository paths as provenance locators. The principal surfaces are:

- `papers/references.bib`, for bibliographic records;
- `source_materials/pdfs/`, for retained source documents;
- `src/mathphysics/`, for reusable implementations;
- `experiments/`, for experiment drivers and retained JSON output;
- `results/`, for retained simulation snapshots; and
- `data/registry/`, for generated indexes and integrity metadata.

Registry presence establishes identity and lineage, not scientific validity. A SHA-256 digest can show that an artifact has not changed; it cannot show that the method that produced it was correct.

1.5 Claim-to-evidence protocol

Each physical claim is evaluated in the same order:

1. state the source claim without strengthening it;
2. identify the standard mathematics or physics it invokes;
3. isolate the extra step needed to reach the claimed conclusion;
4. inspect the repository for an implementation or measurement of that extra step; and
5. record the strongest conclusion the available evidence permits.

This ordering prevents a common failure mode in speculative synthesis: a correct premise is presented, an analogy is inserted, and the final conclusion inherits the authority of the premise despite not following from it.

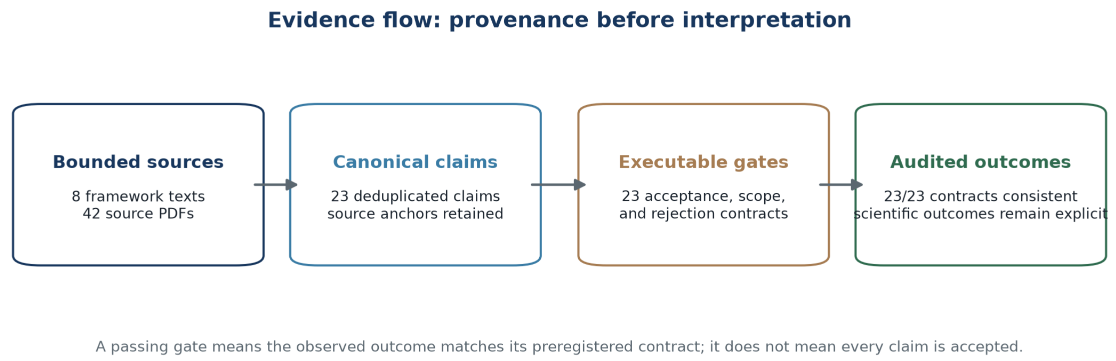

Figure 1.1: The evidence pipeline separates source provenance, canonical claims, executable gates, and admitted conclusions. A passing gate may preserve a falsification or exclusion result.

1.6 Canonical claim disposition

The unified ledger contains 23 claims. Each has exactly one executable gate, and the generated result records whether the observed outcome matches the declared status and rejection contract.

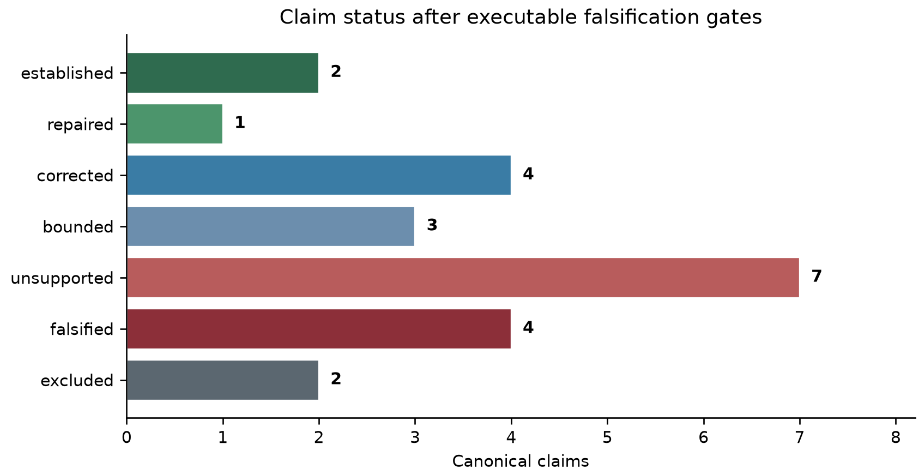

Figure 1.2: Claim statuses after mathematical, computational, scope, and measurability gates.

Table 1.2: Canonical claim status and executable gate outcome.

Claim	Domain	Claim status	Gate outcome
Octonion normed division boundary	Nonassociative algebra	established	accepted
Higher Cayley Dickson physical mapping	Mathematical physics	unsupported	rejected
Albert exceptional group ladder	Jordan algebra	corrected	corrected

Claim	Domain	Claim status	Gate outcome
Spin8 triality	Lie theory	established	accepted
E9 E10 E11 classification	Kac Moody algebra	corrected	corrected
E10 E11 supergravity scope	Quantum gravity	bounded	bounded
Monstrous moonshine stability	Modular forms	unsupported	rejected
Fractional and negative dimension scope	Fractal geometry	corrected	corrected
Fractal estimator convergence	Numerical analysis	bounded	bounded
Scalar field curvature mechanism	Field theory	unsupported	rejected
ZPE energy harvesting	Quantum field theory	unsupported	rejected
Time crystal energy amplification	Condensed matter physics	unsupported	rejected
Planck force unification	Dimensional analysis	unsupported	rejected
Origami fold merge operator	Operator theory	excluded	excluded
E7 root quotient charge	Lattice theory	falsified	falsified
External Fourier quotient map	Lattice theory	falsified	falsified
Kac Moody fluid transfer	Mathematical physics	unsupported	rejected
LBM beta plane and jets	Computational fluid dynamics	falsified	falsified
Matched beta plane algebraic ablation	Computational fluid dynamics	falsified	falsified
LBM equilibrium conservation	Computational fluid dynamics	repaired	repaired
Tourmaline novel energy effect	Materials science	bounded	bounded
Gravitoelectromagnetism scope	General relativity	corrected	corrected
Consciousness resonance	Biophysics	excluded	excluded

1.7 Limits of this review

The review does not independently reproduce every theorem cited in the corpus, nor does it treat repository tests as substitutes for peer review. It evaluates what the checked-in artifacts can support. Claims that require laboratory data, hardware execution, or a new mathematical proof remain open only when the claim is defined precisely enough to test. Otherwise they remain unsupported, not merely unfinished.

The beta-plane study is a short, decaying, periodic-domain implementation test; it does not establish statistically stationary turbulence or persistent jets. The algebra audits combine literature scope with exact counterexamples and finite numerical residuals; sampled identities do not replace universal proofs. No new materials, vacuum-energy, time-crystal-energy, gravitational, or biophysical experiment was performed. Those claims therefore remain bounded, rejected, or excluded according to their admission packages rather than being promoted by the breadth of the source corpus.

CHAPTER 2

Falsification of the Critiques

A critique is itself a claim. This chapter therefore subjects each material objection from the first audit to the same rule applied to the source corpus: name a possible falsifier, inspect the relevant repository surface, and retain only the narrowest verdict supported by the result. A surviving critique is not strengthened by rhetoric. A failed critique is narrowed or withdrawn.

The canonical record is `data/registry/critique_evidence_ledger.json`. The table below is generated from that ledger, and generation fails when an evidence path is missing. This keeps the prose verdict, machine-readable status, and repository evidence on one truth surface.

Table 2.1: Critique-by-critique falsification record.

Critique	Falsification attempt and repository evidence	Verdict
The value c to the fourth divided by G proves force unification	Search the retained sources for field content, a common gauge structure, derived couplings, or a distinguishing prediction. The corpus supplies dimensional identities and interpretations but no such derivation.	Survives: The identity defines a force scale, not a unification mechanism.
Cancellation of hbar is independent physical evidence	Expand Planck energy divided by Planck length. The cancellation is an algebraic consequence of the selected derived units.	Survives: No independent macroscopic quantum effect follows from the cancellation.
E7 roots carry nontrivial P/Q parity	Classify all 126 generated roots. Every root belongs to the root lattice Q and therefore represents the zero coset.	Survives: The stated root-class map is trivial.
No nontrivial Fourier-to-P/Q map can exist	Enumerate every homomorphism from Z^2 to the E7 quotient P/Q isomorphic to Z/2 and impose square-lattice D4 symmetry. The unique nontrivial invariant map is h(kx,ky)=(kx+ky) mod 2.	Falsified: A nontrivial symmetry-compatible map exists, but homomorphism additivity admits every exact triad and therefore supplies no nontrivial triad filter.
A modular triad filter drives the simulation	Search the CPU, JAX, notebook, and high-resolution driver paths for quotient labels and a triad-admission predicate. Neither exists.	Survives: The proposed operator is absent from the implemented dynamics.

Critique	Falsification attempt and repository evidence	Verdict
The retained dynamics implement a beta plane	Trace every parameter into collision, forcing, and streaming. The notebook computes a beta value, but beta does not enter the evolution.	Survives: Arithmetic after the run is not beta-dependent dynamics.
Root enumeration is nondeterministic	Repeat generation across hash seeds and inspect the set-to-array boundary. The tested float tuples were stable, but set iteration was an unstated dependency; the generator now sorts lexicographically.	Repaired: Enumeration is deterministic, while physical invariance under ordering remains unproved.
Root order is physically irrelevant	Compare the canonical density scaffold with controlled permutations. Reversal is exactly invariant because opposite roots share absolute coordinates, while a one-place cyclic shift changes the perturbation by 0.131 in relative L2.	Falsified: One symmetry-related permutation is harmless, but index weighting remains load-bearing.
The high-resolution run has catastrophic mass growth	Read all retained Parquet states. Relative drift reaches only 5.60e-6 at step 500.	Narrowed: Catastrophic growth belonged to the separate historical CPU demonstration.
The historical CPU demonstration conserves mass	Sum the historical D2Q9 equilibrium at nonzero velocity. Its coefficients applied the sound-speed factors twice, producing density times one plus nine times speed squared.	Repaired: The historical critique survives, and the current equation and validator now pass conservation tests.
The retained run supports four zonal jets	Measure the final state with a fixed diagnostic. Mode 17 dominates the small zonal-mean profile, which is only 1.22e-3 of total velocity RMS; the vorticity moment ratio is 0.992.	Survives: A peak index alone does not establish a strong zonal regime.
The reported Rhines comparison is independent	Trace beta_eff in the notebook. It is reconstructed from the observed velocity and peak wavenumber used in the comparison.	Survives: The reported agreement is circular.
Affine symmetry transfers to the fluid through the E8 label	Search for currents, operator products, a stress tensor, and Ward tests. The repository supplies none.	Survives: The Sugawara result does not transfer to the fluid model by naming.
Imaginary algebraic roots are evanescent spatial modes	Search for a dimensional map between the Kac-Moody bilinear form and a spatial dispersion relation. None is defined.	Survives: The identification conflates different quadratic forms.
The corpus contains no empirical materials evidence	Inspect the tourmaline and zero-point-vibration sources. They include ordinary electrical and thermal measurements and standard lattice-vibration calculations.	Falsified: Relevant conventional evidence is present.

Critique	Falsification attempt and repository evidence	Verdict
The corpus supports novel tourmaline or excess-energy effects	Search for calibrated raw data, complete energy balances, blinded controls, and independent replication of the proposed enhancement.	Survives: Conventional evidence does not test the novel coupling.
Two indexed JSON artifacts are unusable	Parse the historical files, identify truncation, and rerun deterministic generators with schema-valid output and invariant tests.	Repaired: The historical critique survives; the current artifacts are parseable and regenerated.
Registry inclusion establishes scientific validity	Compare identity checks with parsing, schema, conservation, and model gates. Registry identity is necessary but not sufficient.	Survives: A digest establishes byte identity, not scientific validity.
The retained fractal dimensions are converged	Compare retained estimates with exact controls and inspect scale-window stability. Multiple estimates disagree and no refinement study closes the gap.	Survives: The values are finite-resolution diagnostics.
Layered harmonics measure an E7 physical effect	Trace the retained schedule to a physical observable or discriminating control. It is a parameter construction without such a link.	Survives: The artifact records a chosen construction rather than a measured E7 effect.
A matched dynamical ablation is the uniquely smallest next test	Separate zero-cost structural checks from dynamical experiments. Exhaustive triad inventory can reject the proposed filter before any simulation.	Falsified: The exhaustive audit closes the homomorphic filter before dynamics; a beta-plane run now serves as a negative matched control rather than a rescue test.

The reassessment changes the paper in three ways. It withdraws absolute claims about the absence of conventional materials evidence, confines catastrophic mass growth to the defective historical CPU run, and records repaired artifacts as current regressions rather than continuing to call them malformed. The main physical objections survive because the missing maps, operators, and controls remain missing after those corrections.

CHAPTER 3

Mathematical Baseline

3.1 Exceptional and extended Lie algebras

The finite-dimensional exceptional complex simple Lie algebras are G_2, F_4, E_6, E_7, and E_8. Their ranks, dimensions, root counts, and Cartan matrices are established classification data. The affine and indefinite extensions commonly denoted E_9, E_{10}, and E_{11} are Kac–Moody algebras, not additional members of the finite exceptional list [Kac, 1990].

For a simply laced finite root system with root lattice Q and weight lattice P, the finite quotient P/Q has order equal to the determinant of the Cartan matrix. Thus

$$|P/Q| = \det A, \qquad \det A_{E_6} = 3, \quad \det A_{E_7} = 2, \quad \det A_{E_8} = 1. \tag{3.1}$$

Evidence status: Established result

These identities classify lattices and representations. They do not, without an explicit mode map and dynamics, determine transport coefficients, forcing amplitudes, jet counts, or material response.

The E_8 lattice is even and unimodular in eight dimensions, and its minimal vectors form the 240 roots of E_8. The lattice also appears in the solution of the eight-dimensional sphere-packing problem [Cohn, 2017]. None of these statements supplies a physical coupling merely by reusing the label E_8 in a simulation.

The exceptional Jordan algebra, or Albert algebra, is the 27-dimensional algebra $J_3(\mathbb{O})$ of Hermitian 3×3 octonionic matrices. Its exceptional group ladder depends on which structure is preserved:

Table 3.1: Corrected exceptional-group roles associated with the Albert algebra.

Group	Dimension	Preserved structure
F_4	52	Jordan-algebra automorphisms
$E_{6(-26)}$	78	determinant-preserving reduced structure
$E_{7(-25)}$	133	conformal/Freudenthal structure

Therefore E_7 is not the automorphism group of a putative $J_3(S)$ built from sedenions. The repository audit now evaluates a nonzero octonionic Jordan product, verifies its identity and commutativity residuals, tests the Jordan identity, and records the three distinct group roles. This replaces the former implementation, whose matrix product returned zero for every input.

3.2 Cayley–Dickson algebras

The Cayley–Dickson construction produces real algebras of dimension 2^n. The first four are

$$\mathbb{R}, \qquad \mathbb{C}, \qquad \mathbb{H}, \qquad \mathbb{O}. \tag{3.2}$$

Algebraic properties are progressively lost under doubling. In particular, the sedenions contain zero divisors, so the existence of a high-dimensional Cayley–Dickson algebra does not make it a division algebra or a viable space of physical states [Baez, 2002, Schafer, 1966].

Evidence status: Established result with a computational illustration

A finite program can exhibit a zero-divisor pair or test identities on sampled elements. Such a run illustrates a known theorem; it neither proves the theorem nor validates a physical use of the algebra.

Table 3.2: Cayley–Dickson property boundary reproduced by the executable audit.

Algebra	Dim.	Comm.	Assoc.	Alt.	Power	Flex.	Norm	ZD
Real	1	Yes	Yes	Yes	Yes	Yes	Yes	No
Complex	2	Yes	Yes	Yes	Yes	Yes	Yes	No
Quaternion	4	No	Yes	Yes	Yes	Yes	Yes	No
Octonion	8	No	No	Yes	Yes	Yes	Yes	No
Sedenion	16	No	No	No	Yes	Yes	No	Yes
Pathion	32	No	No	No	Yes	Yes	No	Yes

Here "Power" means power-associative, "Flex." means flexible, "Norm" means multiplicative norm, and "ZD" records a nonzero zero divisor. The exact sedenion witness is $(e_3+e_{10})(e_6-e_{15})=0$.

3.3 Modular forms and moonshine

A holomorphic modular form of integer weight k for $\mathrm{SL}(2,\mathbb{Z})$ is holomorphic on the upper half-plane, satisfies $f((a\tau+b)/(c\tau+d)) = (c\tau+d)^k f(\tau)$, and is holomorphic at the cusp. For subgroups, every cusp requires its own boundary condition. The modular invariant j is a modular function: it has a pole at the cusp. The classical modular invariant has the expansion

$$j(\tau) = q^{-1} + 744 + 196884q + 21493760q^2 + \cdots, \qquad q = e^{2\pi i\tau}. \tag{3.3}$$

The appearance of Monster-group representation dimensions in coefficients of j led to monstrous moonshine, proved using the Monster Lie algebra and its twisted denominator identities [Conway and Norton, 1979, Borcherds, 1992].

Evidence status: Established result

Recomputing a few coefficients or special values checks an implementation. It does not establish that an unrelated physical system has Monster symmetry. That requires a defined action, observables transforming under it, and tests that distinguish the symmetry from alternatives.

In particular, the theorem identifies graded Monster traces with specified genus-zero modular functions. It supplies no generic stability operator for a fluid, fractal, or material evolution equation. The physical claim is rejected unless a state space, Monster action, evolution operator, invariant or norm, stability inequality, and matched non-moonshine control are all defined.

3.4 Fractal dimensions

Hausdorff and box-counting dimensions are well-defined mathematical invariants, but finite-resolution estimators depend on scale selection, sampling, and regression choices [Falconer, 2004]. A high coefficient of determination for a log–log fit is not by itself evidence of asymptotic scaling. The fitted window must be stable under resolution changes and must avoid the finite-size and discretization plateaus.

3.5 The inference boundary

The established objects in this chapter can parameterize models. They do not select a model on their own. A defensible cross-domain construction must provide all of the following:

1. a map from the mathematical object to physical degrees of freedom;
2. equations of motion that use the mapped structure;
3. units and normalization for every parameter;
4. a baseline model without the proposed structure;
5. an observable that differs between the models; and
6. uncertainty estimates or a convergence study for the comparison.

Without these items, the mathematics may be correct while the physical interpretation remains an unsupported extrapolation.

CHAPTER 4

Audit of the Physical Claims

4.1 Planck force and force unification

The quantity

$$F_{\mathrm{P}} = \frac{c^4}{G} \tag{4.1}$$

has dimensions of force. It can also be written as the Planck energy divided by the Planck length. These are algebraic identities among constants and derived units [Planck, 1899].

Evidence status: Source claim

The reviewed Superforce manuscript interprets this scale as a force of unification [Pais, 2023]. The repository reproduces the dimensional cancellation and the value of the scale.

Evidence status: Assessment: unsupported extrapolation

Dimensional agreement is necessary for an equation but does not derive an interaction. A force-unification claim requires a field content, symmetry, couplings, renormalization behavior, and predictions that differ from general relativity and the Standard Model. None follows from c^4/G alone. Substituting Planck units into Newtonian or Coulomb force laws selects a scale; it does not show that distinct gauge interactions become the same mechanism.

The absence of $\hbar$ from c^4/G also has no independent physical content: $\hbar$ cancels in the ratio of two Planck units. The cancellation cannot, by itself, establish a macroscopic quantum effect.

4.2 Root-lattice charges and fluid triads

The earlier manuscript proposed that Fourier modes inherit a class in the discriminant group P/Q and that triad closure conserves those classes. The proposal fails at its first stated map. If the forcing vectors are roots, then they lie in the root lattice Q and every root represents the zero class in P/Q. The proposed nontrivial parity charge therefore does not arise from the root set as written.

A nontrivial construction would need, at minimum:

1. a homomorphism from the physical Fourier lattice to P/Q;
2. a reason physical modes occupy nonzero weight-lattice cosets;
3. a forcing or interaction operator that enforces the proposed charge;

4. a count of admitted and rejected triads under that operator; and
5. a control forcing with matched spectrum, power, and anisotropy but no quotient structure.

Evidence status: Assessment: mechanism not implemented

The repository contains prose describing a modular filter, but the simulation step contains no P/Q labels and no triad-admission test. The discriminant-group argument cannot explain a numerical result until that missing map and operator exist.

4.3 The beta-plane interpretation

The Rhines scale follows from competition between turbulent advection and Rossby-wave dynamics on a beta plane [Rhines, 1975, Vallis and Maltrud, 1993]. A beta-plane simulation therefore needs a beta-dependent term in its dynamical equations, not merely a spectrum that can be assigned a preferred wavenumber after the run.

The checked-in LBM parameter object has no beta parameter. Its CPU step performs equilibrium evaluation, BGK collision, streaming, and macroscopic update. The accelerated step adds a spatially varying relaxation time and random noise, but no beta-plane body force or potential-vorticity gradient. The high-resolution driver sets grid size, time steps, relaxation time, and harmonic amplitude. It does not configure beta.

The implementation selects the first N roots and weights their absolute first two coordinates by enumeration index. Root generation now uses a deterministic lexicographic order, removing a reproducibility defect. Determinism does not make the construction Weyl invariant. The retained order audit finds that reversal is exactly invariant because lexicographic reversal pairs opposite roots with equal absolute coordinates, while a one-place cyclic shift changes the initialized perturbation by 0.131 in relative L^2. Any observed anisotropy remains confounded with this ordering choice, projection, and index weighting until a symmetry-invariant map and matched controls remove those alternatives.

The earlier diagnostic

$$\beta_{\text{inferred}} = U_\star k_{\text{jet}}^2 \tag{4.2}$$

is useful only after a beta-plane model and an independently defined velocity scale exist. Choosing or reconstructing $U_\star$ from the same output and then declaring agreement with an asserted beta is circular.

The retained final high-resolution snapshot also contradicts the published numbers in the superseded assembly. A direct read of the step-500 snapshot gives a mean speed near 6.92×10^{-3}, a maximum speed near 1.10×10^{-2}, and dominant zonal-mean Fourier mode 17. The zonal-mean RMS is only 8.75×10^{-6}, or 1.22×10^{-3} of total velocity RMS, and the vorticity spectral second-moment ratio is 0.992. Thus the selected peak mode is not evidence of a strong zonal state. The earlier text used $U \approx 0.6$ and mode 4. The run driver does not compute a jet count, spectral slope, or uncertainty interval, so those earlier values have no traceable analysis path.

Table 4.1: Requirements for a defensible beta-plane comparison.

Requirement	Repository state	Consequence
Beta-dependent dynamics	Absent from the retained LBM steps	The run is not a beta-plane experiment.
Independent input beta	No beta field in the parameter object	Input-output agreement cannot be tested.
Prespecified jet diagnostic	Not retained with uncertainty or mode-selection rule	Jet count is post hoc.
Matched controls	No isotropic or spectrum-matched ablation	Algebraic causation is unidentified.
Conservation check	JAX drift is 5.60×10^{-6} at step 500; the repaired CPU regression is conservative	Catastrophic growth does not describe the high-resolution run.

4.4 Affine algebras and conformal invariance

For an affine Lie algebra at level k, the Sugawara construction produces a Virasoro action under specified representation-theoretic conditions. For affine E_8 at level one, the central charge is

$$c = \frac{k \dim E_8}{k + h^\vee} = \frac{248}{31} = 8. \tag{4.3}$$

This is a statement about a current algebra and its conformal field theory. It does not imply that a fluid forced with a pattern labelled E_8 inherits that Virasoro representation. Two-dimensional turbulence can display conformal statistics in appropriate regimes [Bernard et al., 2006], but similarity of vocabulary is not a derivation.

Evidence status: Assessment: category error

Persistent homology can summarize geometry in data. It cannot prove that a fluid field carries a Sugawara stress tensor. The latter requires an explicit current algebra, operator products or equivalent observables, and Ward-identity tests.

4.5 Imaginary roots and evanescent waves

An imaginary root of an indefinite Kac–Moody algebra is defined by the root space bilinear form. It is not an imaginary spatial wavenumber. Identifying the two square roots without a map between the bilinear forms conflates algebraic classification with a dispersion relation. The repository supplies no such map, so the proposed “evanescent threshold” and “superforce limit” are unsupported.

4.6 Gravitoelectromagnetism

Gravitoelectromagnetism is a controlled analogy inside general relativity, not a direct coupling between electromagnetic and gravitational fields. In the linear perturbation construction, a Minkowski background is perturbed by $h_{\mu\nu}$, the trace-reversed perturbation obeys a sourced wave equation after a transverse gauge condition is imposed, and the gravitoelectric and gravitomagnetic fields are defined from the resulting potentials [Mashhoon, 2007]. The Lorentz-like test-body form is therefore conditional on the weak-field expansion, the stated gauge and source definitions, and the retained velocity order.

The analogy does not create a new electromagnetic–gravitational coupling, does not replace the nonlinear Einstein equations, and does not establish a propulsion mechanism. A stronger claim requires a covariant action with the proposed interaction, consistent source and test-body variables, recovery of the general-relativistic limit, and a controlled prediction that ordinary electromagnetic coupling, vibration, and thermal drift do not explain.

4.7 Materials and zero-point-energy claims

The repository retains manuscripts concerning tourmaline, vibrations, and zero-point-energy interpretations. The corpus includes conventional empirical material data, including tourmaline electrical and thermal measurements, and standard first-principles calculations of zero-point vibration. This evidence falsifies an absolute claim that the source collection is wholly nonempirical. It does not test the proposed enhancement, excess-energy effect, or new coupling. Those claims require calibrated raw measurements, a complete energy balance, blinded controls, uncertainty budgets, and independent replication. The checked-in corpus contains no such discriminating evidence.

The experimental-admission registry converts that boundary into operational packages. Measurable proposals receive observables, matched controls, complete energy channels, acceptance criteria, and rejection criteria. Undefined analogies receive no pretend observable and remain excluded.

Table 4.2: Operational content of the physical-claim admission packages.

Claim	Outcome	Obs.	Ctrls.	Channels	Missing
Scalar field curvature mechanism	rejected	3	3	6	4
ZPE energy harvesting	rejected	3	3	8	4
Time crystal energy amplification	rejected	3	3	9	4
Tourmaline novel energy effect	bounded	3	3	9	4
Origami fold merge operator	excluded	0	0	0	4
Consciousness resonance	excluded	0	0	0	4

CHAPTER 5

Computational Evidence Portfolio

5.1 Selection and evaluation rule

This chapter reviews eight retained artifact families. Each entry records the method, the paper-ready output it could support, one validation check, and the strongest permissible inference. This format prevents an implementation check from silently becoming a claim about nature.

Table 5.1: Experiment artifacts, figure targets, and validation gates.

Artifact	Method and figure target	Validation check	Permissible inference
Fractal estimates	`experiments/results/fractal_dimensions.json`. Finite-resolution box counting or correlation sums. Plot fitted slope by scale with the selected window marked.	Repeat at increasing sample counts and shift the fit window. Compare with known dimensions.	Tests the estimator on selected synthetic sets.
Modular forms	`experiments/results/modular_forms_analysis.json`. Truncated series evaluation. Plot absolute error for known special values against truncation order.	Verify $j(i) = 1728$ and the zero at the cubic elliptic point to stated tolerance.	Checks numerical evaluation of standard identities.
Cayley–Dickson	`experiments/results/cayley_dickson_real_validation.json`. Generated property record. A table reports finite identity checks by dimension.	Regenerate atomically, parse as JSON, and compare checks with direct calculations.	Finite computational checks of standard algebraic properties.
E8 analysis	`experiments/results/e8_analysis.json`. Generated root-system summary. Plot root norms and pairwise inner-product multiplicities.	Parse the file, then test root count, rank, norms, opposite closure, and Cartan determinant against independent construction.	Finite validation of the generated E_8 root corpus.

Artifact	Method and figure target	Validation check	Permissible inference
High-resolution LBM Parquet snapshots	BGK-style lattice-Boltzmann evolution with noise. Plot mass, momentum, and kinetic energy versus step before any spectrum.	Inspect schemas, prove fixed boundary and forcing conventions, and bound conservation drift.	Retained numerical trajectories, not beta-plane evidence.
LBM demonstration	`experiments/quantum_lbm_stable_results.json`. A 32 by 32, 50-step conservation regression. Plot total mass and kinetic energy versus step.	Require relative mass drift below a prespecified tolerance.	The corrected CPU path conserves mass; it remains a numerical test, not evidence of quantum or exceptional-algebra dynamics.
Layered harmonics	`experiments/data/genesis_harmonics_e7.json`. Deterministic layered frequencies and amplitudes. Plot amplitude against layer index with the generating formula.	Regenerate byte-equivalent output from a recorded version and seed.	Describes a chosen parameterization; it does not measure E7 physics.
Generated registries	File discovery, size, and SHA-256 capture. Plot artifact counts by kind or use a compact inventory table.	Rebuild registries and verify every digest and schema.	Supports provenance and integrity only.

5.2 Repaired result serialization

The first audit found two experiment JSON files truncated inside scalar values. That historical failure showed that registry inclusion checked identity but not parseability. The current files are generated by `scripts/generate_core_validation_results.py`, written atomically, parsed by tests, and checked against explicit finite invariants. The E8 record now contains 240 roots, 120 positive roots, rank 8, dimension 248, norm-squared multiplicity 240 at value 2, opposite closure, and Cartan determinant 1. The Cayley–Dickson record reports only properties actually calculated by its generator.

Evidence status: Repaired computational artifacts

The files now support their stated finite checks. Repairing serialization does not promote those checks into physical evidence, and future registry generation still needs a general parse-and-schema gate rather than file existence alone.

5.3 Fractal-dimension estimates

The retained fractal result contains scales, measures, fitted dimensions, regression errors, and coefficients of determination. This is a better evidence shape than a single scalar. It still lacks a convergence study over resolution and fit-window selection. For example, the retained Cantor-set estimate is approximately 0.675, while the exact middle-thirds value is $\log 2/\log 3 \approx 0.631$. The discrepancy is large enough that the estimator must be described as finite-resolution and method-dependent.

Evidence status: Computational check

The artifact can support a methods figure showing scale dependence. It cannot support a claim that arbitrary framework geometries have a validated fractal dimension.

5.4 Modular-form special values

The modular-form artifact records $j(i) = 1728$ numerically and a near-zero value at an approximation to the cubic elliptic point. It also records initial coefficients of the j expansion and Ramanujan tau values. These are useful regression targets because independent reference values exist.

Evidence status: Computational check

Agreement verifies a limited numerical implementation. The computation contains no data from a physical system and therefore supplies no evidence for a moonshine-based physical mechanism.

5.5 The CPU LBM defect and repair

The first audit found that the file named `quantum_lbm_stable_results.json` grew from initial mass near 1.024×10^3 to final mass near 1.251×10^6 after 50 steps. The failure was not generic LBM instability. The CPU equilibrium used standard D2Q9 coefficients and then divided by the sound-speed powers a second time. At nonzero velocity its populations summed to $\rho(1 + 9|u|^2)$ rather than ρ.

The equilibrium now uses the dimensionally consistent coefficient form. The conservation validator measures relative error against initialized mass instead of returning a constant verdict. The regenerated 50-step periodic run conserves mass to approximately 1.78×10^{-15} relative error. Tests also inject mass deliberately and require the validator to reject the altered state.

Evidence status: Repaired numerical regression

The corrected result validates one conservation prerequisite for the CPU code. The root-indexed initial condition and passive auxiliary arrays still supply no evidence for quantum transport, coherent flow, or a cross-domain physical map.

5.6 High-resolution snapshot mismatch

The high-resolution driver leaves the requested harmonic count and Reynolds metadata unchanged, giving 127 and 100 respectively. Only 126 E7 roots exist, so the effective initializer count is 126; the Reynolds field does not enter the evolution. The superseded paper assembly instead reported 10 harmonics and Reynolds number 10. A direct read of the final Parquet snapshot finds dominant zonal-mean mode 17, mean speed 6.92×10^{-3}, and maximum speed 1.10×10^{-2}, not the published mode 4 and characteristic speed 0.6. The mode-17 zonal profile is weak: its RMS is 1.22×10^{-3} of total velocity RMS. A vorticity-power second-moment ratio of 0.992 is close to isotropic. Relative mass drift rises smoothly to 5.60×10^{-6} by step 500; this is measurable but not the catastrophic CPU failure. No checked-in analysis computes the reported uncertainty or a $-5/3$ fit.

Evidence status: Retained output contradicts the superseded claim

Every reported table cell must be generated from a versioned analysis command and stored beside the run parameters. Values that cannot be traced to that path are removed rather than rounded into agreement.

5.7 Resolved structural gate and controlled next experiment

The exhaustive triad inventory closes the structural gate. The unique nontrivial square-symmetry-invariant homomorphism from the Fourier lattice to $P/Q \simeq \mathbb{Z}/2$ is $h(k_x, k_y) = k_x + k_y \pmod 2$. It admits every exact triad by additivity, so the proposed filter equals its identity control before any dynamics are run.

The short matched beta-plane ablation tests implementation sensitivity and records the algebraic filter as a prespecified negative control. It is not the scientific jet test. The production program separately executes 540 paired runs across five beta values, three drag values, three viscosities, and twelve independent seed clusters:

1. implement the beta-dependent evolution independently of quotient labels;
2. compare the homomorphic filter with its bit-identical identity control;
3. specify mass, energy, enstrophy, anisotropy, and jet diagnostics before running;
4. execute a committed-before-rerun resolution and time-step refinement study;
5. report seed-clustered effect sizes and simultaneous uncertainty bounds;
6. require zero filter effect to numerical precision and reject any claimed algebraic effect if the trajectories differ.

The numerical, refinement, and control gates pass. The registered scientific hypothesis does not. At $\beta = 5$, the zonal-fraction effect is -0.0030 and the persistent-jet-prevalence effect is 0.1019. At $\beta = 10$, the effects are -0.0200 and 0.0000. Every required simultaneous interval crosses zero, and every required effect estimate misses its locked minimum. A cache-isolated clean-container rerun reproduces the negative aggregate decision. The result is evidence against the declared transient-zonalization effect in this freely decaying model; it is not evidence against beta-plane jet formation under other forcing or dissipation regimes.

This experiment cannot rehabilitate the root-lattice interpretation. The first nonhomomorphic candidate also fails specificity: its E_7 quadratic-defect mask is exactly the generic even-checkerboard kernel at every enumerated triad. Generic parity-kernel dynamics would define a different mechanism and still requires conservation and matched-control gates.

Table 5.2: Required contrasts from the locked beta-plane production matrix.

Metric	Effect	Lower bound	Upper bound	Decision
Zonal-energy fraction at $\beta = 5$	-0.0030	-0.0404	0.0368	Not supported
Persistent-jet prevalence at $\beta = 5$	0.1019	-0.3426	0.5185	Not supported
Zonal-energy fraction at $\beta = 10$	-0.0200	-0.0592	0.0215	Not supported
Persistent-jet prevalence at $\beta = 10$	0.0000	-0.4630	0.4885	Not supported

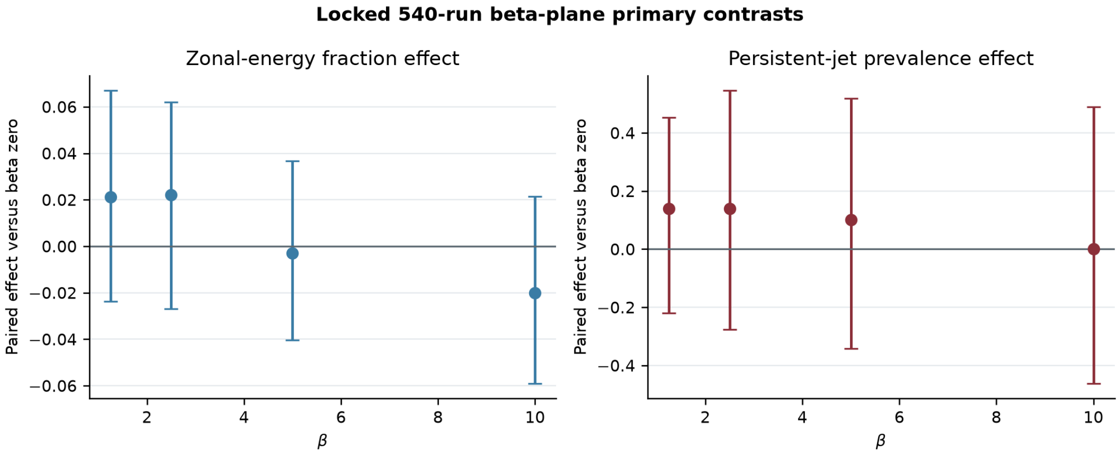

Figure 5.1: Primary effects from the locked 540-run beta-plane matrix. Error bars are simultaneous familywise 95 percent bootstrap intervals over twelve seed clusters. None of the four required $\beta = 5, 10$ contrasts passes.

CHAPTER 6

Reconciled Framework Architecture

6.1 Why concatenation is not integration

The retained framework corpus contains eight text files, but they are not eight independent theories. A deterministic paragraph audit finds that `Alpha001.06` and `Maximal Extraction` contain the same 390 unique normalized blocks. The former repeats these blocks until 2,078 eligible block occurrences remain, of which 1,688 are internal duplicates. Counting these repetitions as agreement would turn copying into evidence.

The consolidation therefore preserves the original documents as archival sources and constructs a new framework from unique claims. Exact duplicate spans collapse to one canonical statement with multiple provenance aliases. Similar headings nominate a semantic review; they do not authorize merging different equations or physical mechanisms.

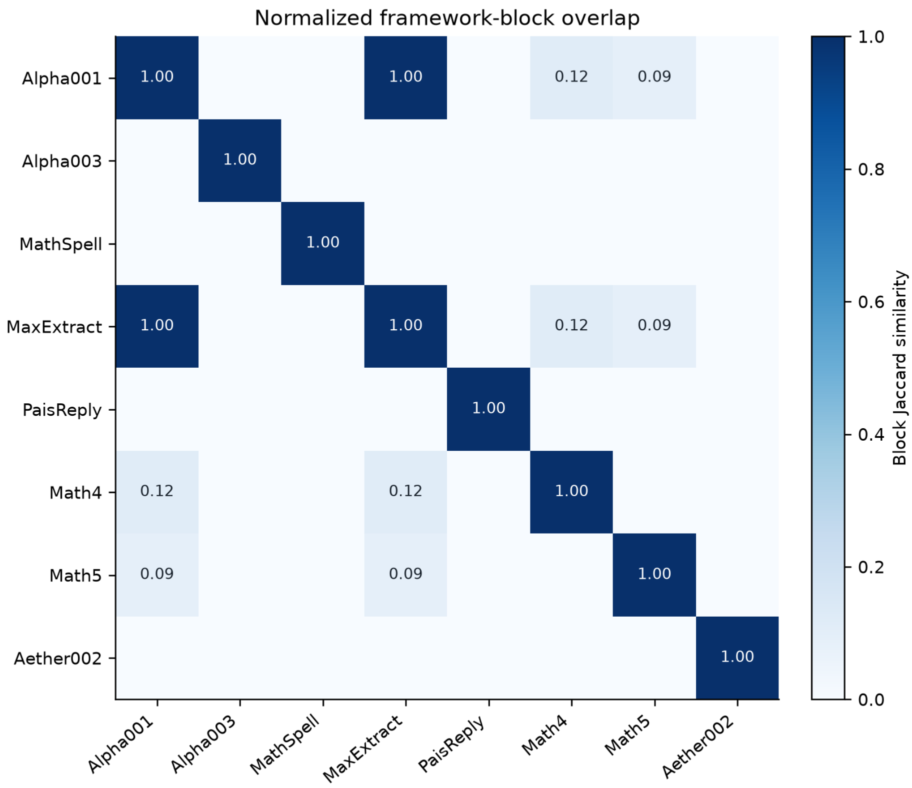

Figure 6.1: Exact normalized-block overlap among the eight retained framework documents. The unit overlap identifies duplicate lineage rather than independent corroboration.

6.2 One framework, five evidence layers

The unified framework is organized by evidential role rather than source order.

Defined mathematics
: Octonions, higher Cayley–Dickson algebras, exceptional and Kac–Moody algebras, the Albert algebra, lattices, modular forms, spectral geometry, and named notions of dimension enter with their standard definitions and known boundaries.

Typed bridges
: A cross-domain map must name its source object, target object, map, preserved invariants, assumptions, and observable.

Reproducible computation
: Code enters only after schema, determinism, invariant, refinement, and control gates appropriate to its claim.

Physical hypotheses
: A proposed mechanism must descend from a defined action, Hamiltonian, constitutive law, or stochastic process and must state units, parameters, observables, and a null model.

Historical ideation
: Undefined analogies and nonempirical claims remain traceable but do not enter the scientific core.

This architecture preserves productive conjectures without allowing them to inherit theorem-level confidence from the mathematical objects they mention.

6.3 Resolved mathematical conflicts

6.3.1 Cayley–Dickson scope

The real normed division algebras stop at the octonions [Baez, 2002]. Higher Cayley–Dickson doublings remain legitimate algebraic objects, and their zero divisors are mathematically structured [Moreno, 2005]; however, division, norm composition, and other identities cannot be assumed to persist. A proposed physical application must identify the exact multiplication law, subspace, invariant, and observable it uses. Dimension alone supplies no mechanism.

6.3.2 Albert algebra and exceptional groups

The source statement that E_6 is the automorphism group of $J_3(\mathbb{O})$ and E_7 the automorphism group of a sedenionic $J_3(\mathbb{S})$ is rejected. The Albert algebra is built from octonions; F_4 is its automorphism group, while suitable forms of E_6 and E_7 arise as structure and conformal groups [Baez, 2002]. These group actions are different objects, not interchangeable meanings of "symmetry".

6.3.3 The E-series boundary

E_6, E_7, and E_8 are finite-dimensional exceptional simple Lie algebras. E_9, E_{10}, and E_{11} are affine, hyperbolic, and very-extended Kac–Moody structures. Restricted supergravity dictionaries motivate research programs [Damour et al., 2002, West, 2001]; they do not prove a complete symmetry of observed interactions.

6.4 Resolved physical boundaries

6.4.1 Time crystals

The equilibrium continuous-time-crystal proposal is excluded under broad conditions by the Watanabe–Oshikawa no-go theorem [Watanabe and Oshikawa, 2015]. Floquet time crystals are instead driven nonequilibrium many-body phases [Else et al., 2016], and primary experiments observe their subharmonic response [Zhang et al., 2017]. The drive, dissipation, and finite lifetime remain part of the energy budget. These results do not supply free amplification of vacuum energy.

6.4.2 Vacuum response and energy extraction

Casimir forces do not by themselves prove that zero-point energy is an extractable reservoir [Jaffe, 2005]. Dynamical-Casimir photon production uses work supplied through a driven boundary condition [Wilson et al., 2011]. Quantum energy inequalities further constrain negative-energy magnitude and duration [Fewster, 2012]. A proposed device must close its complete energy balance over a cycle before an excess- energy claim enters the framework.

6.4.3 Planck force and unification

The quantity c^4/G is a force scale. The cancellation of $\hbar$ in a ratio of derived Planck units is an algebraic identity, not independent physical evidence. A unification claim still requires field content, a common action, coupling relations, correct low-energy limits, and a distinguishing prediction.

6.5 Open hypotheses with admission gates

The root-to-P/Q charge proposal is falsified because every root lies in the root lattice and therefore represents the zero coset. An external map can be defined: the unique nontrivial square-symmetry-invariant homomorphism is $h(k_x, k_y) = k_x + k_y \pmod 2$. It still cannot filter exact convolution triads, because $k + p + q = 0$ implies $h(k) + h(p) + h(q) = 0$. The map-existence question is closed positively and the homomorphic filter mechanism is closed negatively.

The historical fluid evolution contains no beta term and its retained final state is nearly isotropic under fixed diagnostics. The new preregistered barotropic solver includes the beta term and passes budget, refinement, and multi-seed implementation gates. Its homomorphic quotient-filter arm is exactly identical to the identity arm. The short decaying ablation establishes implementation sensitivity only. The separate 540-run production matrix does not support the registered transient-zonalization conjunction; its four required contrasts and simultaneous intervals appear in table 5.2.

A scalar-gravity hypothesis requires a covariant action, stress-energy tensor, background solution, perturbation equations, stability criterion, and an observable distinguishable from an ordinary scalar-tensor model. The generic sourced scalar equation in the drafts does not satisfy this contract.

6.6 Durable authority

The machine-readable authority is `data/registry/unified_framework_claims.json`. It records canonical statements, original source anchors, status, confidence, literature and repository evidence, falsification tests, and integration actions. The human-readable framework is `docs/framework/UNIFIED_EVIDENCE_FRAMEWORK.md`. Unsupported ideas can be reformulated, but a changed claim receives a new identifier rather than erasing the failed version.

CHAPTER 7

Hypothesis Registry and Promotion Gates

The claim ledger and the hypothesis registry serve different purposes. A claim contract checks whether the repository's stated disposition agrees with its retained evidence. It does not turn a rejected or unsupported claim into a successful experiment. The hypothesis registry instead records the formal replacement statement, observable prediction, controls, falsification threshold, required evidence, implementation state, and reproduction state for every bounded, unsupported, excluded, or falsified canonical claim.

The registry contains 14 atomic hypotheses covering 16 canonical source claims. It also dispositions 54 explicit prompts from dormant legacy chapters as mapped, superseded, field-level open problems, or excluded because they are undefined. These archival items are not part of the manuscript's active claim surface. Their retention prevents an old prediction table or open-problem list from silently restoring a claim that the evidence review has already bounded.

7.1 Promotion rule

A supported or negative computational or empirical result enters the paper only after all of the following conditions hold:

1. the hypothesis and its controls are registered before the production run;
2. the implementation passes its numerical and structural gates;
3. the locked decision rule is evaluated without endpoint substitution;
4. a supported claim passes every registered acceptance threshold, while a negative claim reports exactly which component or conjunction fails and does not convert non-support into component-wise exclusion;
5. a cache-isolated pinned environment repeats the decision without reusing the primary run cache;
6. the reproduction record retains the source commit, environment image digest, preregistration digest, primary and reproduction result digests, command, reviewer, and evidence paths; and
7. the hypothesis registry marks the result eligible or promoted.

The same evidence may support a negative conclusion without supporting the proposed mechanism. Exact checkerboard equality falsifies the registered E_7-specificity statement. The beta-plane evidence instead rejects the registered conjunction as not supported; it does not exclude every component effect in isolation.

7.2 Bounded research programs

The program distinction is substantive. The current barotropic model is freely decaying and has no forcing operator or injection-energy budget. Literature on forced–dissipative jet formation constrains interpretation but does not convert this solver into a sustained-jet experiment Bakas et al. [2015, 2019], Sahoo and Ray [2025]. Exact and quasi-resonant Rossby-triad work motivates

Program	Present state	Promotion boundary
Fourier triad selectors	Specificity falsified; reproduced	The failed P/Q homomorphism remains a negative control. The quadratic-defect mask is the generic even-checkerboard kernel. Any parity-kernel experiment still requires conservation, matched-mask, and reproduction gates.
Decaying beta-plane flow	Repeated negative result	The 540-run matrix, controls, and clean rerun agree on the not-supported decision. The result is limited to the registered freely decaying model.
Physical admission	Four operator packages specified	Every physical claim remains blocked by calibration, raw measurements, energy or constraint closure, blinded controls, and independent replication.

Table 7.1: The three active research programs and the evidence boundary that prevents premature manuscript promotion.

equation-derived selector controls without licensing an exceptional-algebra attribution Kartashov and Kartashova [2013], Hayat et al. [2018]. Similarly, the E_7 discriminant form is established mathematics, whereas imposing its quadratic defect as a fluid interaction law is a new conjecture that must beat generic matched masks before any E_7 attribution is possible.

No program passes a positive mechanism claim at this revision. The two cache-isolated negative decisions enter the paper with distinct scopes: exact equality falsifies E_7 specificity, while the beta-plane conjunction is not supported. Neither result becomes evidence for an alternative exceptional-algebra mechanism or for beta-plane behavior outside the registered freely decaying domain. The manuscript otherwise retains the audited mathematical corrections, exact counterexamples, implementation repairs, and explicit physical-evidence boundaries.

CHAPTER 8

Conclusions and Decision Record

8.1 What survives the audit

The repository contains useful mathematical and engineering work when its claims are kept at the scale of the evidence:

- standard exceptional-Lie-algebra invariants provide strong regression targets for implementations;
- Cayley–Dickson computations can illustrate identity loss and zero divisors in finite cases;
- modular-form special values and series coefficients provide independent numerical checks;
- fractal estimators retain enough intermediate data to support a proper scale-window and convergence analysis;
- conventional tourmaline measurements and standard zero-point-vibration calculations provide relevant materials context, although not evidence for the proposed enhancement; and
- registries and digests provide a useful provenance layer for source and result artifacts.

These are defensible contributions. They do not need a unified-field narrative to be valuable.

8.2 What does not survive

The central cross-domain conclusions are not established by the checked-in artifacts:

- c^4/G is a force scale, not a derivation of force unification;
- root vectors have trivial class in P/Q, so the stated discriminant selection rule is vacuous, and the separately defined homomorphic Fourier map admits every exact triad;
- the retained LBM dynamics contain no beta-plane term, so they cannot validate a beta-plane interpretation;
- a Sugawara construction for an affine algebra does not transfer to a fluid field through naming or geometric resemblance;
- imaginary algebraic roots are not imaginary spatial wavenumbers; and
- repairing JSON serialization and CPU mass conservation does not supply the missing physical map, beta dynamics, or matched controls.

8.3 Strongest defensible interpretation

The compendium is best understood as a software-backed review environment for testing mathematical implementations and auditing speculative source claims. It is not evidence for a unified physical framework. The repository becomes more useful when a failed test is allowed to remain a failed test and when a correct identity is not asked to carry a causal conclusion it does not imply.

8.4 Integrated evidence architecture

The review now enforces one directional chain of authority:

1. retained source bytes establish what the reviewed documents contain;
2. native text, MinerU decomposition, and Tesseract fallback provide separately hashed representations of those bytes;
3. the critique ledger binds each objection to a falsification attempt, a controlled verdict vocabulary, and concrete evidence paths;
4. deterministic generators and numerical tests establish finite mathematical or computational properties;
5. schema and digest gates establish that registries describe the live artifacts rather than stale predecessors; and
6. the manuscript states only the narrowest conclusion licensed by all upstream gates.

No downstream layer can promote an upstream result to a stronger evidence class. Better OCR cannot turn a source claim into a measurement. A matching digest cannot turn a malformed model into a valid one. A conservative solver cannot supply a missing physical map. The integration is valuable precisely because each component constrains the others instead of lending them borrowed authority.

8.5 Resolved conjectures and remaining hypotheses

Three initially open conjectures now have negative mechanism results. The unique nontrivial square-symmetry-invariant Fourier homomorphism to E_7's $P/Q \simeq \mathbb{Z}/2$ exists, but additivity admits every exact convolution triad. The proposed nonhomomorphic replacement is exactly the generic even-checkerboard kernel, which falsifies E_7 specificity before a dynamical run. In the 540-run beta-plane program, the required $\beta = 5$ and $\beta = 10$ zonalization and persistent-jet contrasts all miss their locked effects and simultaneous confidence bounds. The historical four-jet conclusion and both algebraic-filter explanations are falsified for their declared implementations. The registered transient-zonalization conjunction is not supported in its parameter domain: required components fail their locked decision rules, although the $\beta = 5$ persistent-jet interval does not exclude its minimum effect in isolation. A pinned clean-container rerun reproduces the selector equality and beta-plane aggregate decision without reusing the primary run cache. These two negative results therefore pass the manuscript promotion gate. They do not support an E_7 fluid mechanism and do not generalize from freely decaying flow to forced–dissipative jet formation.

The remaining physical proposals are not one undifferentiated conjectural framework. The scalar-curvature, vacuum-harvesting, time-crystal-amplification, and tourmaline-novelty packages each define distinct observables, controls, energy channels, and rejection rules. None is admission-ready because the required raw evidence is absent. Fold-merge and consciousness-resonance claims remain excluded because they do not define operational variables. Any revised mechanism receives a new claim identifier and must pass its own gate.

8.6 Publication gate

A future edition may advance a new physical mechanism only when all of the following are present:

1. a complete mathematical definition with typed domains and codomains;

2. a derivation connecting that definition to the implemented equations;
3. a reproducible parameter manifest and parseable result schema;
4. numerical conservation and convergence checks;
5. matched controls or ablations that isolate the claimed mechanism;
6. uncertainty estimates across seeds or experimental replicates; and
7. a falsification criterion fixed before inspecting the result.

The correct immediate outcome is therefore not a stronger unification claim. It is a smaller, testable claim and an experiment capable of returning no.

CHAPTER A

Reproducibility and Artifact Inventory

A.1 Build the manuscript

From the repository root, run:

```
make papers
```

The clean-checkout sequence is:

```
make document-ocr-generate
make document-decomposition-index
```

OCR outputs remain generated local caches under `build/document_ocr/`. The retained run manifest records the container image ID, pinned base image, exact commands, modes, and output roots. The decomposition registry records the hashes needed to compare a regenerated cache with the reviewed run.

The paper build uses the checked-in bibliography and regenerates the registry-driven evidence figures and result tables. The broader scientific and interactive figure suite remains a separate target because it requires the full Python environment and may alter unrelated generated artifacts.

The beta-plane and selector results use a separate pinned clean-room image:

```
EVIDENCE_SOURCE_COMMIT=$(git rev-parse HEAD)
tools/reproduction/build_exact_source_image.sh \
  "$EVIDENCE_SOURCE_COMMIT"
EVIDENCE_SOURCE_COMMIT="$EVIDENCE_SOURCE_COMMIT" \
  docker compose -f tools/reproduction/compose.yaml run --rm
EVIDENCE_SOURCE_COMMIT="$EVIDENCE_SOURCE_COMMIT" \
  make validate-clean-reproduction
```

The builder exports the declared commit with `git archive`; uncommitted host files never enter the build context. The image pins its base by OCI digest, installs the repository lock file, and records a digest manifest for every source-tree file after installation. The reproduction service receives no primary build cache. It writes a separate production result, refinement result, control result, selector audit, raw-array archives, run log, environment report, and host-side image identity under `data/reproduction/`. The clean-reproduction verifier compares the source files inside the image with the declared Git tree, validates the environment manifest, and compares normalized results and raw archives with the primary run. Promotion validation recomputes every declared digest and rejects a reproduction whose runner and reviewer identities match.

The retained clean run executes all 540 production cells, 48 refinement cells, and declared controls with zero resumed cells. Its normalized selector, production, refinement, and control JSON records match the primary outcomes; the three raw-array archives are byte-identical. The verification report binds the exact source tree, image digest, environment, preregistrations, results, archives, run log, and verification harness before either result is admitted.

A.2 Decompose the source documents

The source corpus contains 42 PDFs and 1,076 pages. Embedded text is retained as one evidence layer, not assumed to be correct. The page-level audit routes sparse pages for OCR:

```
python3 scripts/audit_pdf_text_quality.py
```

Formula- and layout-aware extraction uses the pinned MinerU 3.4.4 CUDA image in high-effort hybrid mode:

```
docker compose -f tools/document_ocr/compose.yaml build mineru
docker compose -f tools/document_ocr/compose.yaml run --rm mineru \
  'mineru -p /workspace/source_materials/pdfs \
    -o /workspace/build/document_ocr/mineru_corpus \
    -b hybrid-engine --effort high -m auto \
    --formula true --table true --image-analysis true'
```

Tesseract 5 is an independent fallback, not the primary parser:

```
python3 scripts/run_tesseract_pdf_ocr.py \
  source_materials/pdfs/Comment_on_the_Pais_Superforce_Theory.pdf
```

On the sparse 13-page Superforce commentary, MinerU recovers structured headings, display mathematics, reading order, diagrams, and image regions. Tesseract recovers useful prose but degrades several equations. Native text, MinerU, and Tesseract outputs remain separate so later review can trace which representation supports a quotation or formula.

A.3 Run the repository quality gates

The relevant commands are:

```
make papers
make test
make lint
make verify-offline
```

`make lint-latex` checks generated critique evidence, unresolved stubs, and ASCII source hygiene. `make papers` additionally rebuilds the PDF and fails on unresolved references, duplicate labels, margin overflow, malformed bibliography entries, and build warnings. Python and registry gates remain separate so a missing optional runtime cannot masquerade as success.

A.4 Evidence-bearing artifacts

A.5 Artifact validity rules

An artifact enters a scientific argument only if it passes all applicable checks:

1. the file parses according to its declared format;

Table A.1: Primary repository evidence surfaces used in this review.

Path	Role
`source_materials/pdfs/`	Retained source documents.
`papers/references.bib`	Bibliographic metadata used by this manuscript.
`src/mathphysics/`	Reusable mathematical and numerical implementations.
`experiments/`	Drivers and retained experiment outputs.
`results/`	Parquet simulation snapshots.
`data/registry/artifacts_index.toml`	Generated artifact identities and digests.
`data/registry/experiments_index.toml`	Generated experiment-output inventory.
`data/registry/parquet_audit.json`	Schema-inspection status for retained snapshots.
`data/registry/lbm_evidence_audit.json`	Generated conservation, velocity, zonal, and anisotropy diagnostics.
`data/registry/lbm_root_order_audit.json`	Controlled sensitivity of the root-indexed initializer to ordering.
`data/evidence/beta_plane/`	Deterministic primary production, refinement, and control-array archives.
`data/reproduction/`	Cache-isolated clean-container results, raw arrays, environment record, and logs.
`data/registry/independent_reproduction_registry.json`	Digest-bound promotion decisions for supported and negative results.
`data/registry/critique_evidence_ledger.json`	Canonical status, falsification attempt, verdict, and evidence paths for all 21 critiques.
`data/registry/pdf_text_quality_audit.json`	Page-level embedded-text quality and OCR routing.
`data/registry/document_decomposition_audit.json`	MinerU output hashes and artifact counts for all 42 PDFs and 1,076 pages.
`data/registry/document_ocr_run_manifest.json`	MinerU container identity, exact modes and commands, and Tesseract manifest identity.
`data/registry/pdf_ocr_comparison.json`	Native, MinerU, and Tesseract comparison for the sparsest document.
`data/registry/pdf_ocr_visual_review.json`	Retained page-level equation, layout, diagram, and prose comparison rubric.
`data/registry/claim_source_crosswalk.toml`	Source lineage for selected claims.

2. its schema includes units, parameters, and version or commit identity;
3. its digest matches the generated registry;
4. the producing command is recorded and succeeds from a clean environment;
5. prerequisite invariants, including conservation or normalization, pass;
6. the analysis method and selection rules are specified before the result;
7. the claimed inference does not exceed what the artifact measures.

Checksums satisfy only the third item. A registry should not label an artifact scientifically valid merely because it exists and hashes reproducibly.

A.6 Current limitations and blockers

At the time of this review, serialization, CPU mass conservation, deterministic root enumeration, Cartan construction, Albert multiplication, PDF decomposition, and Parquet schema inspection have been repaired. The structural Fourier map and matched beta-plane questions have executable negative results. The remaining scientific blockers are:

- the historical high-resolution LBM driver has no beta-plane parameter or term and remains evidence only for its own nearly isotropic trajectory;
- the homomorphic Fourier-to-P/Q mechanism is closed negatively, and the first nonhomomorphic candidate is exactly a generic checkerboard kernel;
- the high-resolution states lack a complete retained parameter and software-version manifest;
- the reported four-jet and $-5/3$ conclusions have no traceable analysis;
- the 540-run freely decaying beta-plane matrix does not support its registered transient-zonalization effect within the tested domain and does not license inference outside it; and
- scalar gravity, ZPE harvesting, time-crystal amplification, and tourmaline novelty lack the raw measurements required by their operational admission packages.

These blockers are evidence, not housekeeping. They determine which claims the manuscript can honestly make.

Declarations

Document type. This work is a repository-backed review monograph. A venue-specific journal template is outside the evidence claim and must not change the registered results.

Author contributions. Eirikr Hinngart curated the source corpus, implemented the audits and experiments, reviewed the evidence, and prepared the manuscript. Automated tools assisted code generation, OCR, testing, and adversarial review; their activity remains attributable through the repository history and does not constitute authorship.

Data and code availability. Source, code, registries, and retained evidence are available at `https://github.com/Oichkatzelesfrettschen/MathScienceCompendium`. The pre-experiment evidence baseline is tag `v0.1.0-evidence`. Exact source, container, protocol, result, archive, and verification digests for the promoted negative results are recorded in `data/registry/independent_reproduction_registry.json`.

Competing interests. The author declares no competing interests in the claims evaluated in this review.

Ethics. This review reports no new human-participant or animal study.

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